COVID Mortality Disruption Lab

How the pandemic impacted non-COVID mortality

Research Question: Did COVID-19 disrupt the trend for six other leading causes of death, has it recovered, and where did it hit hardest?

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Summary

I tested six causes of death against their own pre-pandemic trends using CDC WONDER mortality data from 1999 to 2024. **Five of them moved significantly off trend in 2020-21**, and five of those results hold up after correcting for testing six causes at once. Most of that damage has since faded. By 2024, only diabetes (+15.0%) and cancer (+4.6%) were still significantly above trend; heart disease, stroke, and drug overdose were back within their expected range.

Two results went against what I predicted before looking at the data. Cancer was expected to show nothing in this time window and showed a small but real increase. Alzheimer's disease was expected to show a large increase and showed none in 2020-21, though a decline appears in later years. A negative control, three sensitivity checks, and a multiple-testing correction were all run before any result was reported.

COVID-19 itself is not what's being studied here. It's treated as a shock to the health system, and the question is what happened to deaths from other causes afterward. Mortality data can show what changed, by how much, and where. It can't show why, because direct infection, delayed care, isolation, and economic stress all leave the same mark in the numbers. Every result below is an association, not a causal claim.

This kind of excess-mortality analysis is how public health agencies estimate a pandemic's toll beyond deaths officially attributed to the virus: missed cancer screenings, interrupted addiction treatment, delayed heart care. Each cause got a written prediction and confidence level before any 2020-2024 data was pulled, so the results can be checked against what was expected rather than explained after the fact.

1. Research question

> Which causes of death were most disrupted during the COVID-19 pandemic (2020-2024), how long did those disruptions last, and how did their size vary across U.S. counties by income, health insurance coverage, and rurality?

The six causes tested are heart disease, diabetes, Alzheimer's disease, stroke, drug overdose, and cancer. Two more series fill out the design: congenital malformations as a negative control (a cause with no direct link to COVID), and COVID-19 deaths as a reference that isn't itself tested.

2. Methods

Excess mortality with a fixed breakpoint. For each cause, I fit a straight-line trend to its pre-pandemic age-adjusted death rate and projected it forward through 2020-2024 with a 95% prediction interval. A year counts as disrupted when the observed rate falls outside that interval. Most causes use a 1999-2019 baseline; heart disease and stroke use 2010-2019 for reasons covered in section 4. The breakpoint is 2020 because that's when the pandemic started, not because a search found it.

How long it lasted. Causes with a significant 2020-21 disruption are sorted into three groups based on the years 2022-2024. *Persisted* means the later years that were still significantly off trend leaned the same way as 2020-21. It doesn't require the most recent year to be off trend. *Resolved* means no later year was significantly off trend, and *Reversed* means the later significant years leaned the opposite way.

Cross-check. Three change-point methods (PELT, binary segmentation, and segmented regression) were run on each series to see whether they find a break near 2020 on their own. They search the whole series for the single best breakpoint, so landing somewhere else doesn't count against a result at the fixed 2020 date.

Negative control. The same pipeline was run on a cause with no direct COVID link. If it had shown a significant disruption, the method would be picking up an artifact, and no other result could be trusted.

Multiple testing. Testing six causes at once makes a chance "significant" result more likely, so the six p-values are adjusted together with the Benjamini-Hochberg false discovery rate (FDR) method. The county analysis gets its own correction for each cause.

Full-period check. Each cause also gets a second p-value that pools all five post-2020 years instead of just 2020-21. It doesn't replace the main test, since averaging more years can hide a reversal as easily as it can reveal a slow effect, but it catches disruptions that showed up after 2021 (see Alzheimer's in section 3).

Autocorrelation check. The main test assumes each baseline year's deviation from trend is independent of the last. For several causes it isn't, so the same test is also run with Newey-West (HAC) standard errors, which allow for that (section 4).

Data. CDC WONDER Underlying Cause of Death, from two databases joined at their 2018-2019 overlap: D76 (1999-2020) for the baseline and D158 (2018-2024, Single Race) for 2020-2024. County data for diabetes and drug overdose compares 2015-2019 with 2020-2024 and is matched to County Health Rankings & Roadmaps measures.

3. Results

Figure 1 shows the method on drug overdose, the cause with the largest jump. The solid line is the observed death rate. The dashed gray line is where the 1999-2019 trend says the rate should have gone, and the two thin pink lines mark its 95% prediction interval. Overdose deaths broke above that interval in 2020 and stayed there through 2023, then fell back in 2024.

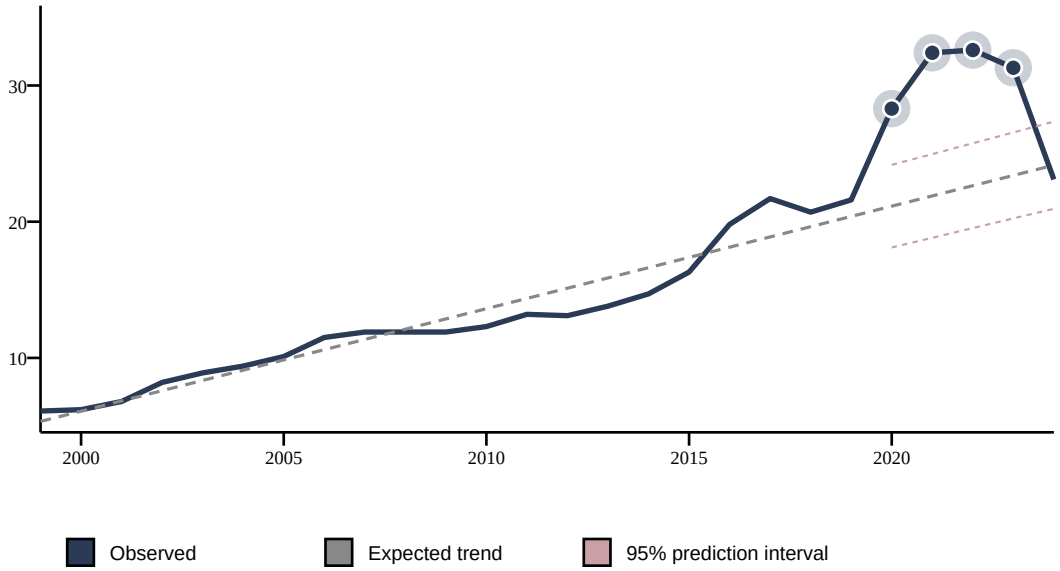

Figure 1. Drug overdose: observed death rate against its pre-pandemic trend, 1999-2024.

5 of 6 causes show a significant 2020-21 disruption, and **5 of 6** hold up after FDR correction.

Cause	Result	p-value	FDR-sig.	2020-21 dev.	2024 dev.	Robust?
Drug overdose	Persisted	5.98e-08	Yes	+40.9%	-4.3%	Yes
Diabetes	Persisted	7.99e-07	Yes	+26.9%	+15.0%	Yes
Heart disease	Persisted	3.86e-05	Yes	+7.8%	+3.0%	Yes
Cancer	Persisted	0.000194	Yes	+1.7%	+4.6%	Yes
Stroke	Persisted	0.00169	Yes	+8.8%	+6.1%	No
Alzheimer's disease	No significant disruption	0.807	No	+1.0%	-19.1%	Yes

"Dev." is the effect size: how far the observed rate sits from the expected trend, in percent. A result can be significant and still small. "Robust?" shows whether the result survives a curved baseline in place of a straight one; only stroke doesn't (section 4).

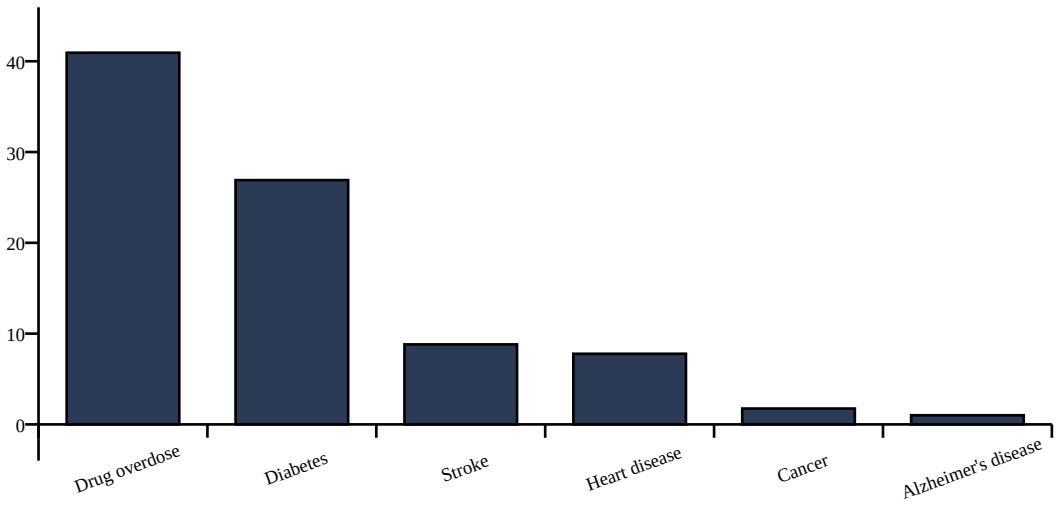

Figure 2. Average 2020-21 deviation from the expected trend, by cause.

Figure 3 repeats Figure 1 for all six causes, with each cause's p-value in its panel title. It's worth noticing how poorly the size of a gap predicts significance. Cancer's gap is barely visible but significant, because its year-to-year noise before 2020 was so small. The next two sections explain why some results hold up and others need caution.

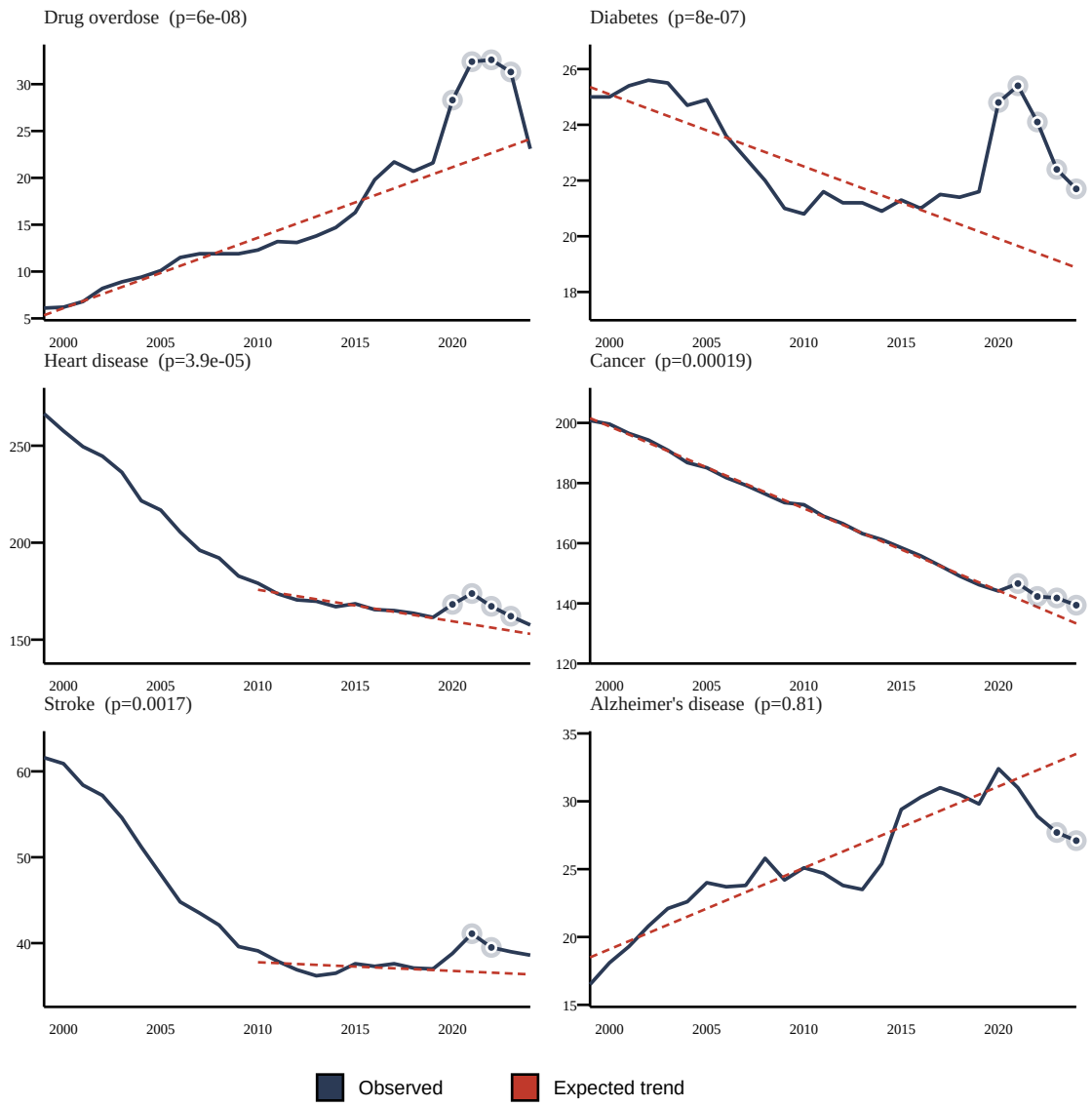

Figure 3. All six causes, observed vs. expected trend, 1999-2024.

Two results that went against the predictions

Cancer was expected to show nothing. The prediction was a null result, since missed screenings were thought to take longer than this data window to show up as deaths. Instead cancer shows a significant disruption (p = 0.000194, survives FDR correction), though a small one: +1.7% in 2020-21 and +4.6% in 2024. Other research doesn't settle this either. Early models predicted large increases in cancer deaths from delayed screening, while later work using English data found lung and breast cancer deaths came in below pre-pandemic trends.

Alzheimer's was expected to show a large effect. The reasoning was that isolation and disruption in care facilities would show up clearly in dementia deaths. It didn't (p = 0.81 for 2020-21). That doesn't rule out an effect, only one visible in national death rates with this method. Pooling all five post-2020 years does find a later decline (p = 0.0019, averaging -8.7% vs. trend). One likely explanation is mortality displacement: people who would have died in 2023-2024 may have died earlier in the pandemic. The main classification still uses the pre-registered 2020-21 window, so this is reported as an additional finding.

Negative control

Congenital malformations, a cause concentrated in infancy with no direct COVID link, shows no significant disruption, as expected (p = 0.682 on raw death counts). Its age-adjusted *rate* does drift +14% above trend by 2024, which is why the gate uses raw death counts: WONDER reports this rate to one decimal place at about 3 per 100,000, and the rate-based test is too sensitive to that rounding. Passing doesn't prove every other result is real, but failing would have been strong evidence of a problem.

The first negative control, accidental drowning, failed. Drowning deaths really did rise from 2020 on, in line with CDC reports tied to pool and beach closures and lifeguard shortages. It was replaced because it turned out to be affected by the pandemic, not because the method broke. The replacement isn't perfectly insulated either, since prenatal and obstetric care were also disrupted in 2020-21, but that path is smaller and less direct than for the six test causes.

4. Robustness

A baseline correction for heart disease and stroke. Both originally used the same 1999-2019 baseline as the other causes, and both failed the trend-shape check below. Using only 1999-2019 data, an F-test comparing straight-line and curved fits showed strong curvature for these two causes (F=222.7 and F=162.6, both p<0.00001) and not for the others (F<9.5). Both fell steeply through the 2000s and then leveled off, so a straight line across the whole period already sat 17.7 and 5.8 points below the real 2019 rate before the pandemic. A curved fit matches history well but predicts *rising* rates after 2019, a common problem when extending a polynomial past its data, so it wasn't used. A straight line over 2010-2019 has neither problem and is now the baseline for these two causes. Both stay significant, but their estimated 2020-21 deviation drops from 26-37% to 8-9%. Details are in research_protocol.md (2026-09-01 addendum).

With that correction in place, three sensitivity checks re-run the main test with one modeling choice changed at a time.

Baseline window (1999-2019 vs. 2010-2019). No result changes.

Significance threshold (α=0.05 vs. α=0.01). One cause changes: Stroke.

Baseline shape (straight vs. curved). One cause changes: Stroke.

The baseline-shape check matters most. Heart disease now passes it and stays significant under either a straight or curved baseline. Stroke improves but doesn't fully pass: its p-value under a curved baseline went from 0.37 with the old baseline to 0.096 with the corrected one, still above 0.05. That makes stroke the least certain "Persisted" result in this report. Diabetes, drug overdose, and cancer pass every check.

Lag-1 autocorrelation measures whether a year that runs above trend tends to be followed by another one. In the pre-pandemic baselines it is high for diabetes, overdose, and Alzheimer's (0.65-0.82), moderate for stroke (0.50), and low for heart disease and cancer (0.12-0.19). Heart disease and stroke had much higher values (0.92 and 0.93) on the old 1999-2019 baseline, which is another sign that a straight line was the wrong fit there. Because the standard test assumes no autocorrelation, the same test was rerun with Newey-West (HAC) standard errors. That raises the p-values for diabetes, overdose, and stroke by one to two orders of magnitude, but all 5 significant causes stay significant.

Cause	Autocorrelation	Classical p-value	HAC p-value
Drug overdose	0.75	5.98e-08	2.53e-05

Heart disease	0.12	3.86e-05	2.7e-05
Diabetes	0.82	7.99e-07	6.66e-05
Cancer	0.19	0.000194	0.000659
Stroke	0.50	0.00169	0.00205
Alzheimer's disease	0.65	0.807	0.857

Joining the two databases. In the 2018-2019 years both databases cover, every cause matched within the 10% reliability threshold, and the median difference was 0% for every cause.

5. Which counties were hit hardest

County data exists for two causes, diabetes and drug overdose. For each county, disruption is the change in average death rate from 2015-2019 to 2020-2024, which is then compared against County Health Rankings & Roadmaps measures. Only 926 of about 3,143 U.S. counties qualify for diabetes and 641 for drug overdose. CDC WONDER hides any county-year with too few deaths to protect privacy, and a county needs at least two reported years in each period to be included. County rates are crude rather than age-adjusted, because WONDER doesn't offer age adjustment at the county level for its 2018-2024 database, so some of a county's change could come from its population getting older.

For diabetes, **Grant County, IN** had the largest increase (60.2 to 123.0 per 100,000), while **Butler County, OH** barely changed (+0.0). For drug overdose, **Mercer County, WV** had the largest increase (57.0 to 119.9 per 100,000), while **Washington Parish, LA** barely changed (-0.1). This analysis has no data on local policies or on age and race, so it doesn't try to explain any single county. A single county's averages are also noisy, since a few extra deaths in a small population can move them a lot.

Diabetes: 5 of 5 measures hold up after FDR correction

County measure	Slope	p-value	FDR-sig.
% adult obesity	21.030	0.000545	Yes
Median household income (per $10k)	-0.544	0.000886	Yes
% adult smokers	24.239	0.00244	Yes
% rural	-3.614	0.00566	Yes
% uninsured	13.709	0.0128	Yes

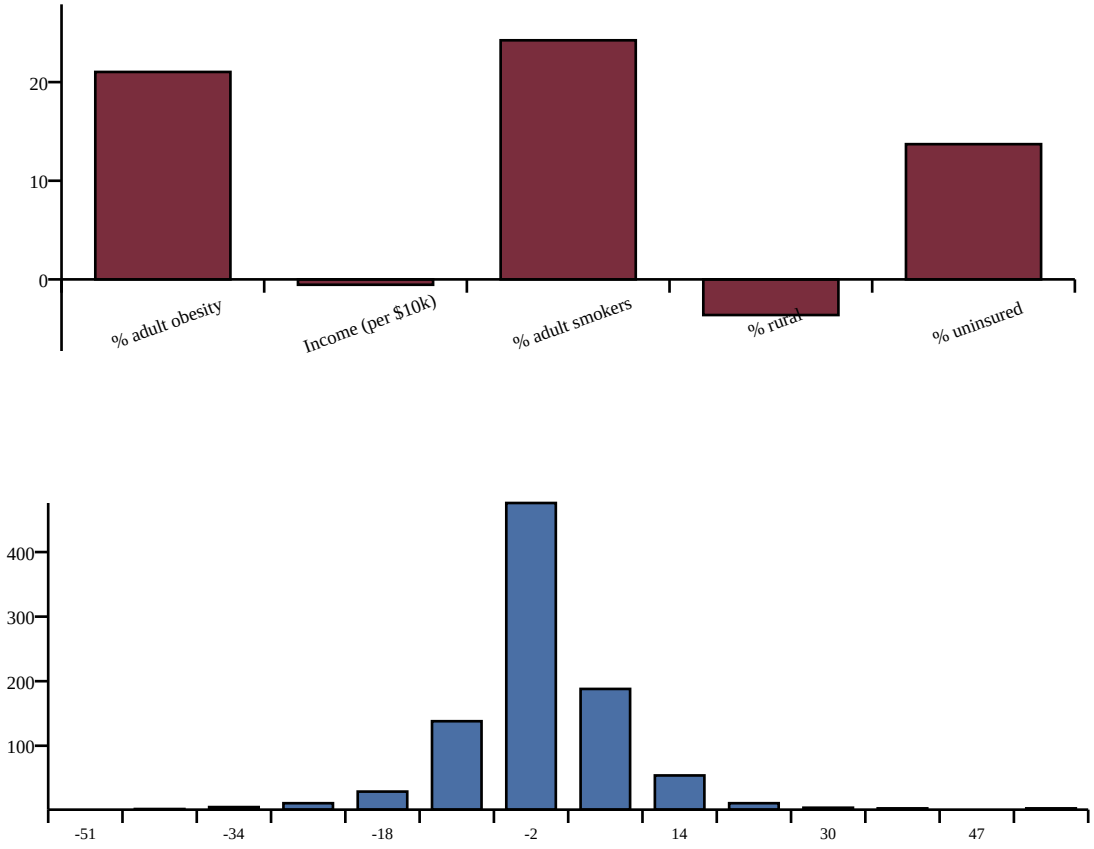

Figure 4. Diabetes: how each county measure relates to disruption (top), and how disruption was spread across the 926 included counties (bottom; disruption on the x-axis, number of counties on the y-axis).

Drug overdose: 4 of 5 measures hold up after FDR correction

County measure	Slope	p-value	FDR-sig.
Median household income (per $10k)	-1.725	3.68e-15	Yes
% adult smokers	68.159	8.36e-09	Yes
% uninsured	38.233	4.32e-06	Yes
% adult obesity	24.497	0.00409	Yes
% rural	-3.942	0.0688	No

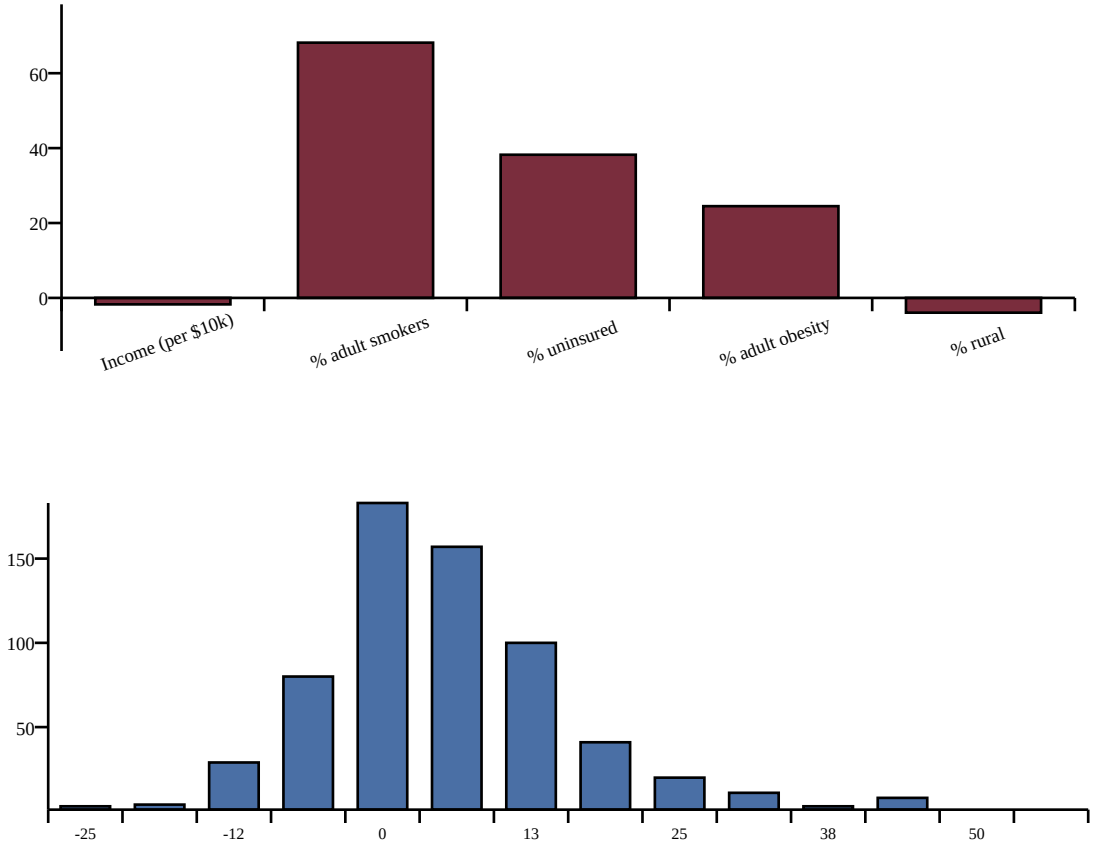

Figure 5. Drug overdose: how each county measure relates to disruption (top), and how disruption was spread across the 641 included counties (bottom; disruption on the x-axis, number of counties on the y-axis).

For both causes, counties with higher uninsured, smoking, and obesity rates saw larger disruption, and higher-income counties saw smaller disruption. More rural counties saw smaller disruption for diabetes; for drug overdose the same pattern doesn't survive FDR correction. That runs against the common assumption that rural areas were hit hardest, and the caveat below matters for reading it. These are associations, not causes.

A caveat on the rurality result

The counties left out for having too few reported years are much more rural than the ones included (for diabetes, excluded counties average 78% rural vs. 31% for included ones; for drug overdose, excluded counties average 74% rural vs. 24% for included ones). So this analysis mostly describes suburban and small-city counties. Splitting the included counties at their median rurality shows how much each result can be trusted:

Diabetes: the pattern holds in the more rural half ($p=0.00325$) and is stronger there than in the less rural half ($p=0.176$).

Drug overdose: the pattern comes almost entirely from the less rural half ($p=0.00016$) and isn't significant in the more rural half ($p=0.4$), so treat it with real caution.

6. Limitations

• This is an observational study of population averages, so it can't support causal claims about individuals. At the county level there's also a risk of ecological fallacy: a pattern across counties may not hold for the people living in them.

• Counties dropped for missing data are mostly rural, which limits what the county results say about rural America. The rurality pattern holds up better for diabetes than for drug overdose (section 5).

• The two CDC WONDER databases may not line up perfectly. The 2018-2019 overlap check reduces this risk but can't remove it.

• Death certificate coding may have shifted in 2020-21 while reporting systems were strained, separate from any real change in deaths.

• Cancer was predicted to show no disruption and showed a small one. The cause isn't clear, and its +4.6% deviation in 2024 is modest.

• The standard test assumes baseline years are independent, which isn't true for several causes (autocorrelation 0.65-0.82 for diabetes, overdose, and Alzheimer's; 0.50 for stroke; 0.12-0.19 for heart disease and cancer). The HAC version of the test accounts for this, and all 5 significant causes stay significant under it (section 4).

• Significant doesn't mean large. Drug overdose and diabetes had the biggest 2020-21 jumps (roughly 27-41%), while cancer, heart disease, and stroke were real but much smaller (about 2-9%).

• Heart disease and stroke use a 2010-2019 baseline instead of 1999-2019 because the longer window didn't fit their pre-pandemic trend (section 4). Heart disease now holds up under either baseline shape. Stroke remains the least certain result.

• County rates are crude, not age-adjusted, so some measured disruption may reflect each county's population getting older.

• Small counties are suppressed or unstable in the county data.

• Smoking and obesity rates come from modeled small-area estimates (PLACES), not direct counts.

• The county measures come from the 2024 County Health Rankings release, so they describe counties during or after the pandemic, not before it. Income and insurance rates may have changed during the pandemic itself.

• Many tests were run: across the six causes, and across the county measures for each cause. Both sets are corrected separately.

• Neighboring counties aren't independent of each other, and this analysis doesn't model that spatial correlation.

• Mortality data alone can't separate direct infection, delayed care, isolation, and economic stress. Every finding here is an association.

7. Data and reproducibility

National results use 15 CDC WONDER exports (7 from the 1999-2020 database, 8 from the 2018-2024 database). County results use 4 more (diabetes and drug overdose, before and after 2020). No synthetic data is used anywhere. The full method, the predictions made before seeing the data, and every change made along the way are in research_protocol.md. Export steps are in manual_data_acquisition.md. The interactive app and source code are at github.com/aidanc667/covid-mortality-disruption-lab.

This project uses public aggregate data for research and education. It isn't medical advice and doesn't predict risk for any individual.